

protection to ensure long life, solar inverter and DC energy meter that tells the user how much solar energy he has used at any given moment.

System development and installation in India

Everything is made in India, except the basic components. But the quality of most of these products is poor. There are only a few reputed Indian manufacturers.

So far, the technology and its solutions have demonstrated their potential in meeting the energy demand. However, as the policies of the central and the state governments as well as the market are still evolving, India has a long road ahead towards self-sufficiency.

Installation costs of rooftop PV systems vary widely depending on capacity, location, quality and brand of the components.

"In the absence of any standardisation and third-party audit of the installations, solution providers are under pressure to be competitive in both controlled and open market. The market today is going through a vicious 'learning cycle' wherein new inexperienced players promise attractive prices for systems that may fail to perform as expected over the next 25 years. However, as with any new market, the market for solar PV systems too will mature over the next few years," says Goel.

"Any well-designed rooftop solar system will cost about Rs 125 to 150 per Watt-peak. However, it is not just about the price but choosing the right source. If I were a buyer I would doubt anything cheaper. One should get subsidy on this price. The decision to choose a supplier only because it is MNRE-approved also needs caution, because I have seen that what is approved and what is supplied are often different," shares Deodhar.

Steps to streamline and standardise solar installations

"The MNRE has introduced a system of inducting channel partners

Grid-interactive vs non-grid-interactive systems

The capacities of diesel generators used in various industries and commercial establishments and housing complexes in cities and towns for backup power, even during the day time, varies from a few kilowatts to a couple of megawatts. Generally, in a single establishment more than one generator is installed—one to cater to the minimum load required for lighting and using computer and other emergency operations during load-shedding and the others for running ACs, lifts etc.

With an objective to reduce dependency on diesel gensets, the Indian government is proposing a scheme to replace them with solar photovoltaic (SPV). Further, in order to utilise the existing roof space of buildings, the scheme proposes to promote rooftop SPV systems on buildings to replace DG gensets installed for minimum load requirement for operation during load-shedding. These loads generally vary between 25 kW and 100 kW or so.

A rooftop SPV system could be with or without grid interaction. In grid-interactive system, the DC power generated from SPV panels is converted into AC power using a power-conditioning unit and fed to the grid, either of 11kV three-phase line or of 220V single-phase line, depending on the system installed at an institution/commercial establishment or residential complex. Rooftop SPV systems generate power during the day time, which is utilised fully by powering the captive loads and feeding excess power to the grid as long as the grid is available.

In cases where solar power is not sufficient due to cloud cover, etc, the captive loads are served by drawing power from the grid. The grid-interactive rooftop SPV systems thus work on net metering basis wherein the beneficiary pays to the utility on net meter reading basis only. Ideally, grid-interactive systems do not require battery backup as the grid acts as the backup for feeding excess solar power and vice versa.

However, to enhance the performance reliability of the overall systems, a minimum battery backup of one hour of load capacity is strongly recommended. In grid-interactive systems, it has, however, to be ensured that in case the grid fails, the solar power has to be fully utilised or stopped from immediately feeding to the grid so as to safeguard any grid person/technician from getting electrocuted while working on the grid for maintenance, etc. This feature is termed as 'islanding protection.'

Non-grid interactive systems ideally require a full-load-capacity battery power backup system. However, with the introduction of advanced load management and power conditioning systems, and safety mechanisms, it is possible to segregate the day time loads to be served directly by solar power without necessarily going through the battery backup. As in the previous case of grid-interactive systems, minimum one hour of battery backup is, however, strongly recommended for these systems also to enhance the performance reliability of the systems.

Non-grid-interactive systems with minimum battery backup are viable only at places where normal power is not available during day time. In case the SPV power is to be used after sunshine hours, it would require full-load-capacity battery backup. This will increase the cost of system, which may not be economically viable even with support from the government.

—Source: MNRE website

after evaluating both the technical as well as the financial strength of the system/solution providers. This is the first step towards introducing an essence of accountability with the system provider. The MNRE has also defined benchmark costs of PV installations, based on which subsidy on capital equipment is extended to the user. However, the route to adopting

PV systems is far from user-friendly and highly time-consuming from the point of view of the beneficiary," informs Goel.

Area: A limiting factor?

In the absence of a solar rooftop policy in the country, whereby one can export electricity into the local distribution network and earn monetary returns,